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Article

# Demand-Side Opportunities for Process Industry Participation in Electricity Markets

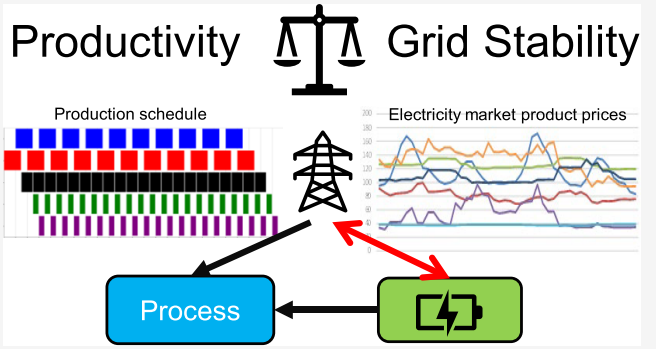
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**ABSTRACT:** This paper elaborates on the topic of industrial demand side management, which has been a matter of research for about two decades. The "traditional" demand side management approach is based on estimating the future energy need and trying to align the load with the energy price forecast in order to minimize electricity cost. For power-intensive industries, this can have a significant economic impact. Here, we investigate the option of using an energy storage unit to make the energy consumption, i.e., the industrial load, more flexible. Installing a battery energy storage system can open up new opportunities, and we include the power reserve markets in the demand side management scope to allow large industrial loads to help keep the power grid stable.




## 1. INTRODUCTION

Several energy-intensive industries have been investigating the consequences and opportunities of electrification, for instance, air industries, steel plants, and cement plants.[1−3] Electrification strengthens the coupling with the power grids, and any disturbances in energy availability will more directly impact the production. Equivalently, the effect of process disturbances will impose a larger stress on the electricity infrastructure, thus calling for approaches to identify and realize possible synergies. As it is in everyone's interest to have a stable electricity supply, industrial demand side management (DSM) has provided more transparency and incentives for collaboration.[4] However, in plain terms, the collaboration between industrial consumers and power grids is still limited to two major schemes:

1. Controllable loads, where the grid operator can reduce/increase the load within specified limits. Examples of this are, e.g., large commercial building HVAC systems and some aluminum smelters in the United States (Alcoa),[3] as well as growing number of data centers.
2. Day-ahead bidding, where the industrial production process defines its production schedule based on an electricity price forecast and availability, trying to minimize its electricity cost while meeting the production targets.[5]

Both approaches have their merits and limitations. Controllable loads are fully managed by the network operator, and while, e.g., the ALCOA Warrick site reports enabling around 70 MW for this purpose,[3] the remuneration mechanism is still largely based on standard generator schemes. Thus, possible disruptions in production processes (the main business of the company) are not fully considered, and the magnitude of the regulation support is not reflected, i.e., the price per MW is the same for 10 MW as for 50 MW. However, operationally, these two situations can be significantly different. On the other hand, day-ahead bidding is purely based on price signals (or forecasts), and this is a relatively stiff approach that leaves little short-term flexibility for the production process. Unplanned disturbances in processes might disrupt the planned load schedule, resulting in heavy penalties that undermine the whole purpose of participating in demand response programs. Also, in many industrial environments, other base tariffs dominate, making DSM rather marginal, as the potential benefits are lower than the potential instabilities caused by price tracking. For instance, interrupting or delaying a smelting process step might result in temperature losses and the need for reprocessing, increasing the total energy need.

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The topic of DSM first gained interest during the 1970s oil crisis, when rocketing energy prices created an urgent incentive for energy conservation and management.[6] Before the crisis, energy availability had been taken for granted, but suddenly, it became necessary to give high priority to the development of methods to better manage energy use. Although interest in DSM was reduced after the crisis, time-of-use and real-time pricing schemes were introduced to incentivize timing consumption for times of less congestion. The traditional consumption model was restored, and power and energy availability were again assumed to follow demand, and industrial processes were scheduled based on other constraining factors.

Process scheduling developed later with the increase in computational power from something done manually to finding mathematical optima, using computational models such as continuous-time approaches,[7] state-task network[8] (STN), and resource-task network[9] (RTN) models. The models are widely used for minimizing factors influencing process costs, such as makespan[10] and cycle time,[11] while considering the availability of other process variables, such as equipment, material, utilities, and personnel, and other resources when needed. Early DSM-adoption of the RTN model was done by Castro et al.[11−13] who expanded it to fit the needs of cement, steel, and the pulp and paper industry scheduling problems. DSM was again introduced as part of process scheduling when cost reductions for power-intensive processes realized their potential. Ierapetritou et al.[14] developed an MILP-based two-stage stochastic programming approach for minimizing the costs of an air separation plant based on electricity prices.

The potential for reducing electricity costs was recognized for many industrial processes, most notably air separation,[15−17] steel making industry,[13,18−20] electrolytic processes,[21,22] cement production,[23] and other heavy process industries.[24−26] With the increased share of renewable energy, questions about maintaining power balance and power availability have become a key factor for scheduling processes alongside cost minimization.[27] DSM and demand response are challenging to implement for many industrial processes, and keeping up with production demand is a higher priority. Additionally, considering larger systems through enterprise-wide optimization (EWO) has suggested benefits in reducing energy costs and facilitating deeper renewables penetration.[28,29]

Similarly, implementing additional energy or intermediate product storage has been suggested to enable DSM capability for processes that previously have been deemed too inflexible. Zhang et al.[17] studied the increased DSM possibilities when implementing cryogenic product storage for liquid oxygen and nitrogen in an air separation plant and found that plants with lower utilization rates gained a significant reduction in costs, up to 20%, decreasing with higher utilization rates. Tiggeloven et al.[27] studied the electrification of the naphtha cracking process for ethylene production and analyzed a case with differing battery sizes, stating that due to high investment cost, little benefit was achieved, although increased process flexibility offered a significant cost reduction. Similar findings on process flexibility were reported by Otashu and Baldea[26] and Roh et al.[30] For battery energy storage systems, the limitations in battery capacities and their high cost limit the economic feasibility for process industries due to high power and energy demands.[31] The variations in hourly price markets are therefore too fast for many processes to exploit and too slow for short-term storage, such as batteries.

Additionally, frequent rescheduling to accommodate demand response targets might increase schedule "nervousness," i.e., negative impact on the shop floor due to frequent schedule changes. Frequent changes in operational plans break routine tasks and may lead to reduced efficiency, motivation, and increased frustration among the plant floor personnel and costs in material and equipment planning and thereby limit the flexibility of the process.[32−36]

More recently, there has been interest in industrial demand response, e.g., providing grid balancing with industrial loads.[29,37−39] The main challenges identified are related to price and production uncertainty and optimal multimarket participation.[40] Germscheid et al.[41] showed, using a biobjective stochastic programming approach, minimizing value-at-risk and expected cost, that participation in both day-ahead and intraday markets can significantly reduce costs compared to day-ahead markets, for both flexible and inflexible processes. Machine learning (ML) models have shown promising results for multimarket participation[42,43] and production scheduling,[44,45] especially in online settings where mathematical models might be too slow. In this paper, we extend the conventional DSM concept by investigating the added flexibility provided by the installation of a battery energy storage system (BESS) analyzing the potential benefits.[46] After BESS installation, we also investigate the opportunities to participate in the power reserve markets in an ideal setting.

## 2. PROBLEM DESCRIPTION

DSM has proven its potential benefits in selected energy-intensive industries. Nonetheless, it has not been utilized as much as could be expected, regardless of the increased energy costs and many changes within the energy sector. One of the reasons is the lack of incentives, and building a reliable DSM system requires both investments and process changes that will disturb the normal day-to-day production at least for a limited time. The efforts so far have been very industry- or even company-specific, and implementing industrial DSM in a new environment will also impact the current planning system and production environment. If no clear proof of a return on investment exists for a company, it will likely abandon or postpone any ambitions in this direction.

In this paper, we discuss various levels of awareness of electricity prices (see Figure 1 for an illustration). Example A

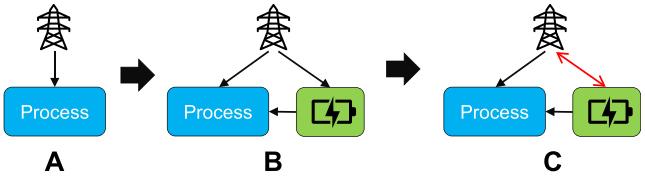



**Figure 1.** Three levels of process and energy interaction.

on the left-hand side reflects a typical one-way flow of electricity to the process. Here, the process can be planned with or without taking into account future electricity prices, typically through price forecast information. In earlier times, the industry commonly had contracts with fixed electricity prices, which could also contain different prices for the day and night, but most often, the production schedules were generated focusing on the process efficiency and customer orders. For

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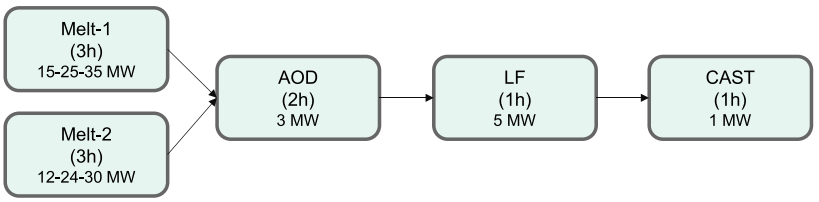



**Figure 2.** Example four-stage energy-intensive process, inspired by the steel making process.

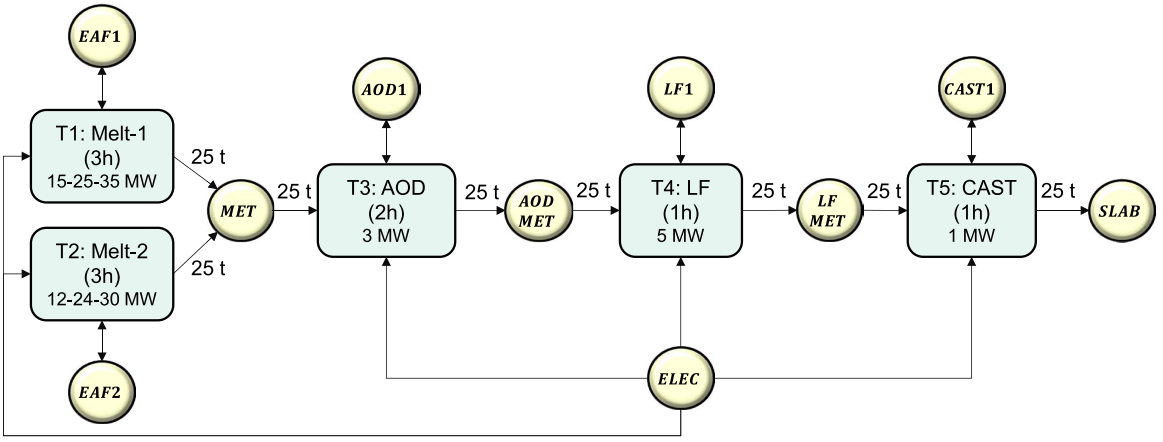



**Figure 3.** RTN graph of the simplified steel production process.

instance, one can simply minimize the total production time (makespan (MS)) to lower the personnel cost and maximize the throughput. With the increasing portion of renewable energy sources (RES) and thus larger volatility in electricity generation, the price differences have become higher, and at some point, adapting the schedules according to the most reliable price forecast in A stands for a reason.

In this paper, we investigate the above options of A, i.e., optimizing the process with and without considering the electricity price. Option B couples the industrial process with an energy storage unit, typically a BESS, which can buffer electricity during lower price periods and feed it to the process during price peaks. This enables a stronger capability to optimally plan the energy procurement, even without disrupting the production process. In the following, we will investigate the impact of the BESS size on the total electricity cost and also try to define the optimal BESS size.

If a company has invested in a BESS, there are multiple ways to monetize it. One of them is to use it for offering ancillary services, e.g., in the form of primary reserves to the power grid and supporting the grid stability. In most locations, reserve provisions need to be certified, i.e., the providing entity must prove its capability to respond if it is called upon during an ancillary event. Renewable energy systems, for example, do not often qualify here, and production processes would also need to prove their flexibility, leading to investments at minimum in the automation system. However, a BESS can often directly qualify for ancillary services as long as it is operated in a way that ensures the availability of the required energy. This means, in practice, that in addition to option B and option C, the process owner can sell back energy to the grid—often into the power reserve markets, which can be a lucrative option to help lower the total electricity cost.[47]

In order to elaborate on these options, we use a steel plant-like example process adapted from Harjunkoski and Grossmann,[10] resulting in a reduced and relaxed steel plant scheduling model. It has two parallel electric arc furnaces (EAFs), followed by three other processing stages, as shown in Figure 2. Having a simplified process allows us to focus on the conceptual changes and not to discuss all the practical process issues since scheduling a steel plant is, in general, a very specific and complex problem.[48]

The melting process is here modeled as a 3 h process with different hourly melting stages requiring different power levels. The longer processing time is used since we only have access to hourly real electricity price information and want to consider the typical phases of the steel smelting. Figure 2 shows the process durations and the corresponding electricity consumption. The electricity demand of the melting process increases along the processing (e.g., 15 MW for hour 1, 25 MW for hour 2, and 35 MW for hour 3 in Melt-1). In the second AOD stage, we have a 2 h constant consumption of 3 MW and the two final processing stages (LF and CAST) both last 1 h. We assume a constant batch size of 25 t across the entire process without any material losses. This example shows a strongly simplified small steel production process, ignoring the cooling and energy losses between subsequent process stages. To efficiently capture the electricity aspects, we model the process using the discrete-time RTN approach[9] presented in Castro et al.,[19] which considered demand response aspects of a cement plant. By representing all resources by circles and tasks by rectangles, we can build the RTN graph in Figure 3. The resources are electric arc furnaces (EAF1/2), argon–oxygen decarburization unit (AOD), ladle furnace (LF), continuous caster (CAST), and electricity (ELEC). In the RTN graph, we must also track all intermediate products as they feed into subsequent processing steps. We name these according to the last processing step: molten metal (MET), which becomes after AOD (AODMET) and after LF (LFMET). The resource exiting the continuous casting step is the final product (SLAB).

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Table 1. Electricity Price (EUR/MWh) for 28−29.6.2023

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 95.39 | 97.60 | 107.08 | 128.92 | 155.18 | 167.62 | 159.10 | 140.00 | 121.10 | 119.46 | 110.06 | 101.72 | 98.08 | 99.92 | 104.52 | 121.81 | 130.55 | 130.31 | 116.39 | 105.95 | 98.60 | 99.99 | 98.24 | 98.60 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 100.00 | 101.12 | 112.03 | 131.26 | 166.25 | 171.58 | 154.73 | 141.81 | 121.11 | 115.53 | 103.40 | 96.55 | 96.34 | 94.75 | 105.34 | 123.71 | 135.64 | 132.92 | 123.28 | 107.22 | 100.76 | 98.44 | 86.27 | 83.15 |

We use this example process to investigate the three alternative interactions with the power grid, as shown in Figure 1.

## 3. MATHEMATICAL FORMULATION AND EXAMPLES

There exist multiple approaches to modelling a process with energy constraints, e.g., as done with continuous-time formulations for steel plants[20] or using other approaches,[14] but this often results in long solution times requiring decomposition approaches.[49] In this paper, we want to focus on the potential of a stronger energy integration and the RTN approach[9,11] offers good tools for this. A comparison between RTN continuous- and discrete-time approaches[19] concluded that the latter can be expected to be more efficient in this context. Using the RTN diagram as a source, we can easily build the corresponding mathematical model for optimizing the process. The general RTN-structure is presented shortly in the following. Assume a given number of processing tasks $i$, a set of discrete-time points $t$, and a binary variable, $N_{i,t}$, indicating whether task $i$ starts at time point $t$. Further assume a continuous variable $R_{r,t}$, tracking the amount (tons) of resources $r$ at each time point $t$, and variable $\xi_{i,t}$ to denote the batch size (tons) of task $i$ starting at time point $t$. Using these variables, we can define the most important resource balance equation (eq 1), which considers the initial resource $r$ amount $R_r^0$ at initial time point 1, else referring to the previous time point $R_{r,t-1}$.

$$R_{r,t} = R_r^0|_{t=1} + R_{r,t-1} + \sum_{i\in I}\sum_{\theta=0}^{\tau_i} (\mu_{i,r,\theta} N_{i,t-\theta} + v_{i,r,\theta}\xi_{i,t-\theta}) + \pi_{r,t} \quad \forall\, r, t \tag{1}$$

Here, for simplicity, as well as due to the hourly electricity price information, we use a time discretization of 1 h. Thus, the actual processing time is equal to the number of time intervals ($\tau_i$) needed for processing the task. The third term with the double sum expresses the resource consumption or production of each task. The parameter $\mu_{i,r,\theta}$ (tons) tracks the absolute amount of resource $r$ consumed (negative parameter) or produced (positive parameter) at each time point $\theta$ relative to the starting time of the task. If the resource amount is relative to the selected batch size $\xi_{i,t}$, the parameter $v_{i,r,\theta}$ defines this relationship. Note that the latter parameter has values between −1 and +1, and in the most common case, either $\mu_{i,r,\theta}$ or $v_{i,r,\theta}$ is used per resource. The final variable, $\pi_{r,t}$, indicates external interaction with the environment, e.g., in this case, electricity purchase or sales (MWh). Each resource must stay within given bounds (tons), as shown in eq 2:

$$R_{r,t}^{\min} \le R_{r,t} \le R_{r,t}^{\max} \quad \forall\, r \in R, t \in T \tag{2}$$

For electricity, the lower and upper bounds are zero, unless we have energy storage units available, since all electricity must be purchased at the time of use ($\pi_{r,t}$). The batch size is limited and defined by the starting time of the task, as shown in eq 3:

$$\xi_i^{\min} \cdot N_{i,t} \le \xi_{i,t} \le \xi_i^{\max} \cdot N_{i,t} \quad \forall\, r \in R, t \in T \tag{3}$$

Finally, variable definitions are shown in eq 4:

$$\pi_t \in \mathbb{R},\ R_{r,t},\ \xi_{i,t} \in \mathbb{R}^+,\ N_{i,t} \in \{0, 1\} \tag{4}$$

In this paper, we assume predefined batch sizes and define only absolute resource amounts, eliminating the variable $\xi_{i,t}$ from the problem, i.e., the second term in the sum of eq 1 and entire eq 3.

**3.1. Optimization of A-Cases.** In the first case, A1, we optimize the production without considering the electricity price at all—as well as without having any energy storage available. Specifically, we only minimize the total production time, i.e., the MS, which is always higher than the start-time $t$ plus duration $\tau_i$ of any task $i$, as shown in eq 5.

$$\mathrm{MS} \ge N_{i,t} \cdot (t + \tau_i - 1) \ \forall\, i, t \tag{5}$$

The corresponding objective function is given in eq 6.

$$\min \mathrm{MS} \tag{6}$$

As noted, this simple objective function is completely ignorant of any electricity pricing and thus more typical for earlier industrial environments where the energy price was fixed and not time-dependent. For considering varying electricity pricing, $c_{\mathrm{elec},t}$ (day-ahead market), the natural way is to add the variable $\pi_{r,t}$, expressing an external interaction, into the objective function. It could be weighed with the MS, but a more realistic approach is to use a profit function where the revenues of the produced slabs are combined with the electricity cost. As the example problem has been scaled down (to about 25% of the common size), and the processing tasks have also been stretched in time (to about 3−4 times longer than usual), in order to have a task span across multiple price intervals, we can only state that our example has been inspired by the steel melt-shop process. Nonetheless, we will use a typical steel price of 2500 EUR/ton, assuming that, except for the energy, the production costs (labor, equipment, etc.) constitute about 70% of the product value. For this, we define a profit factor, $p_{\mathrm{fact}}$, which will thus have a value of 0.3. Also, here we assume that one batch results in one slab of 25 tons, whereas a batch size (heat) in steel making is typically around 100−130 tons. To ensure similar results, we assume a fixed demand of 525 tons (21 batches) across 2 days (48 h). Thus, it is more important to focus on the relative differences than on absolute values. The objective function maximizing the profit is shown in eq 7, where the profit factor is multiplied by the steel price and the produced amount at the last time point, deducted by the electricity cost across all time points. Please note that since the demand and thus revenue are fixed, the objective practically reduces to minimizing the cost.

$$\max p_{\mathrm{fact}} \cdot c_{\mathrm{slab}} \cdot R_{\mathrm{slab},|t|} - \sum_t c_{\mathrm{elec},t} \cdot \pi_{\mathrm{elec},t} \tag{7}$$

This is already a trivial DSM approach considering the dynamically changing electricity prices. The two objectives in eqs 6 and 7 are compared using the electricity prices given in Table 1. The prices are day-ahead energy prices taken from the Danish DK-2 (https://en.energinet.dk/) spot market (June 28−29th, 2023), indicating cheaper prices with green, higher

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with yellow, and most expensive prices with red color. The prices are rather high, and therefore, we could expect to observe a difference. Due to the scope of this paper where we analyze the theoretically maximum potential of adapting production to varying electricity pricing, we will not consider price uncertainty but assume a perfect electricity price forecast.

When optimizing the two above-described problems, which are defined as,

- A1 (minimize makespan): Eqs 1−5 and objective 6.
- A2 (maximize profit): Eqs 1−4 and objective 7.

We obtain the results shown in Table 2.

**Table 2. Summary of the Results A1 and A2**

| problem | profit | electricity cost | makespan |
|---|---|---|---|
| A1 | 185,757.1 | 207,992.9 | 47 |
| A2 | 202,088.5 | 191,661.5 | 48 |

From the results, we can conclude that optimizing the MS of this downscaled example process produces 525 tons of steel (21 batches) one hour faster, but with about 8% higher electricity cost. The electricity cost directly affects the profit. The corresponding Gantt charts are shown in Figure 4. In the chart, it is clearly shown that an MS optimized process is very regular, and the AOD (unit AOF1) is the bottleneck. Changing to profit maximization, where the electricity price is critical, results in a few changes: (a) EAF1, which consumes more electricity, is used only when necessary and not at all during the red, expensive-priced time slots, (b) as the electricity is cheapest at the end of the process, the MS is prolonged by the unused hour, and (c) the entire process is focused on the cheaper hours, making it less organized from the plant floor perspective.

This shows that optimizing the MS of this downscaled example process produces a very even and operationally efficient schedule, whereas doing the cost minimization using the objective function in eq 7 results in a much less organized schedule. It would be practically infeasible if we were to consider all real-world constraints for steel making and require intermediate storage capacity, as well as offer fewer buffers to balance for disturbances. Thus, the schedule on the right-hand side of Figure 4 is mainly theoretical, showing the maximum effect of the cost optimization. Nonetheless, it should be noted that we only allow the existence of one intermediate batch at a time. Adjusting the production based on the energy price is challenging and makes a generic approach difficult to implement on the plant floor. A working DSM solution requires more process-specific adaptations to ensure and secure the necessary process-specific conditions.

### 3.2. Optimization of B-Cases by Adding BESS.

A strong coupling between the production process and electricity prices may not result in practical solutions. Thus, it is of interest to investigate the decoupling of these using a BESS as a buffer between the electricity markets and the process, as depicted in case B of Figure 1. In this case, we add a BESS into the system, and the electricity can be purchased either directly to power the process or to charge a battery, which can then later be discharged to the process. The key question is how large a capacity would be required? In the latter plan (A2), the maximum load is 68.5 MW, and the total energy consumed is 1711.5 MWh.

To test different battery sizes, we run the same simulations using the objective function in eq 7 but adapting the upper bound $R_{r,t}^{\max}$ for the resource electricity accordingly. For a more comprehensive approach, BESS can also be fully embedded into the market model. For fairness, we assume that the BESS is empty (0 MWh) at the beginning in order not to provide any additional hidden benefits for the comparison. The battery capacities tested are 10, 20, 30, 40, and 50 MWh, and the results of the runs are shown in Table 3. Logically, we can see

**Table 3. Cost Optimization with Different BESS Capacities**

| BESS size (MWh) | profit | electricity cost | profit improvement |
|---|---|---|---|
| 10 | 203,657.98 | 190,092.02 | 0.8% |
| 20 | 205,243.48 | 188,506.52 | 1.6% |
| 30 | 208,347.40 | 185,402.60 | 3.1% |
| 40 | 209,368.60 | 184,381.40 | 3.6% |
| 50 | 212,013.67 | 181,736.33 | 4.9% |
| 1373.5 (free) | 286,041.77 | 107,708.09 | 41.5% |
| 44.5 (incl. CAPEX) | 202,458.33 | 182,715.19 | 0.2% |

that the profit is improved by 4.9% through lower electricity costs with a 50 MWh BESS, compared to 0 MW (case A2). If we let the optimization decide on the best storage size (assuming a BESS is completely free), we get the battery storage size of 1373.5 MWh. This is obviously not a plausible result but, on the one hand, gives us the maximum savings potential that can be achieved through using a BESS, which is a profit increase of 41.5% compared to having no BESS at hand (case A2). However, what if the CAPEX investment price is considered? We do a very quick and rough cost estimate. According to most information available on the Internet (https://www.lzyess.com/news/649.html, accessed 2025-08-10), for a BESS system, one needs to count on 200−450 kUSD per MWh, heavily depending on the dis-/charging (MW) capability. Let us here assume an investment cost of 250 kEUR/MWh, 5% interest rate, and 10-year lifetime, during which the BESS must be paid back. Considering annual

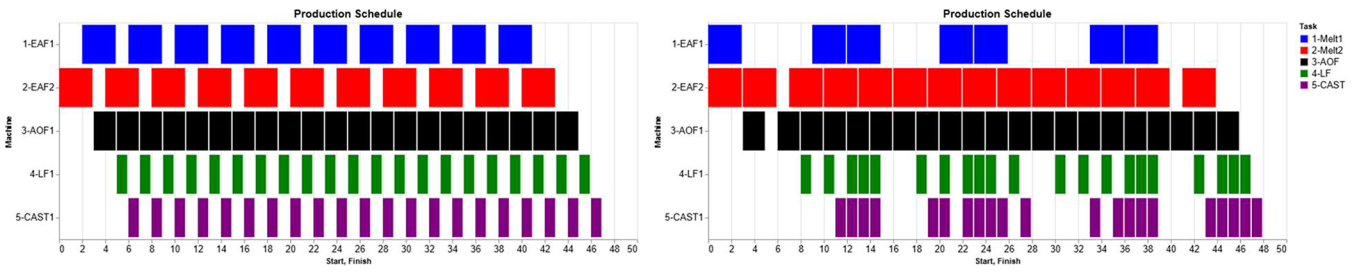



**Figure 4.** Gantt charts for the A1 (left) and A2 (right) cases (hours).

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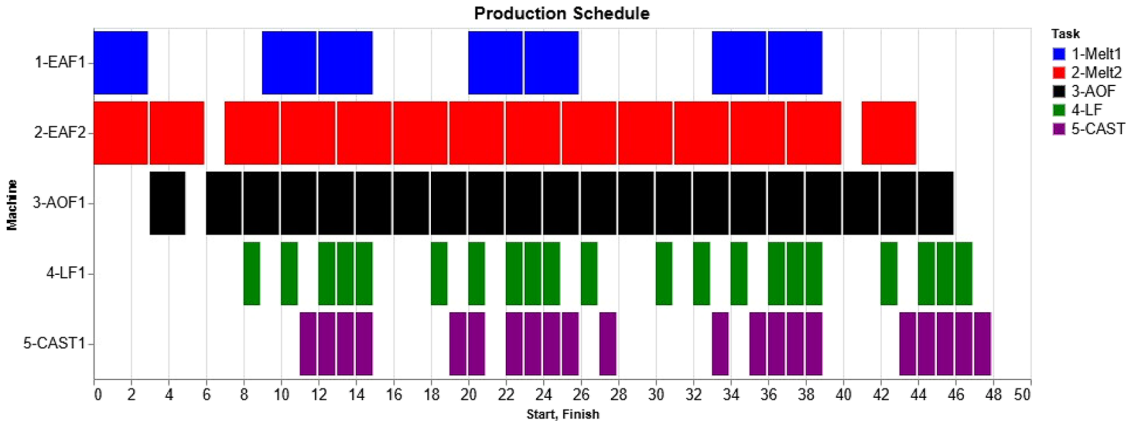



**Figure 5.** Gantt chart using a the BESS of 44.5 MWh (time unit: hours).

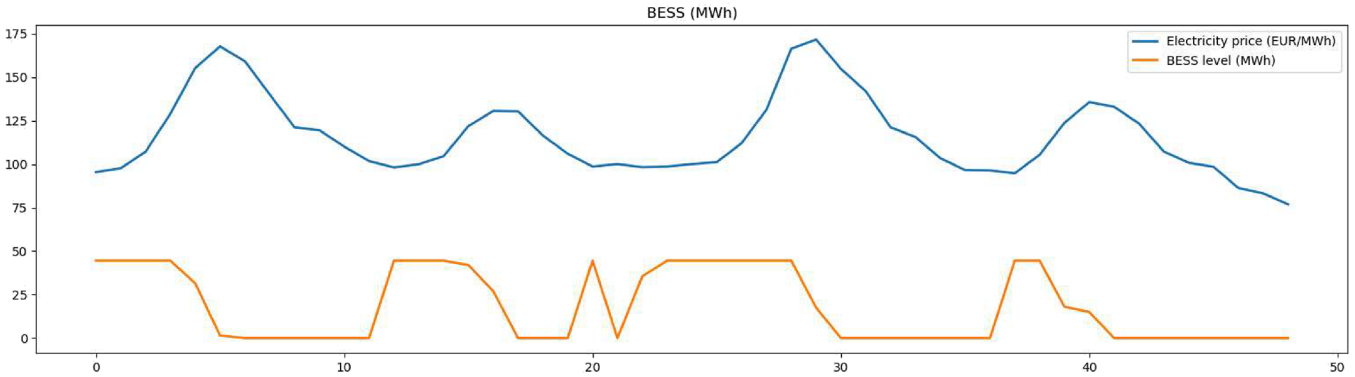



**Figure 6.** BESS used for the battery size of 44.5 MWh (units shown in legend).

payments of 35.1725 kEUR/MWh gives an equivalent of 192.73 EUR/MWh for the 48 h study period. This number is used as the cost coefficient for the investment, which at least gives a realistic limitation on the investment. We add the CAPEX contribution, $c_{MW}$, for the BESS (EUR/MWh) and the 48 h period, as well as a positive variable, $B^{size}$, for the battery storage size (MWh) and update the objective function as in eq 8. The resource limit, $R^{max}_{r,t}$, for electricity in eq 2 is replaced by $B^{size}$:

$$\max p_{fact} \cdot c_{slab} \cdot R_{slab,|t|} - \sum_{t} c_{elec,t} \cdot \pi_{elec,t} - c_{MW} \cdot B^{size} \quad (8)$$

The resulting optimal battery storage size is 44.5 MWh with the CAPEX cost of 8576.49 EUR for the 48-h horizon. As can be seen in Table 3, summarizing all the above results, the profit improvement compared to case A2 is only 0.2% if the investment cost is included.

In this study, battery degradation was not explicitly considered within the RTN model. However, as will be later shown in the paper, the energy purchasing can also be handled by the integrated market model (see eq 10), which considers both dis/charging efficiency factors of 0.95, as well as a cost per MWh for dis/charging activities. We used this link for the cases with a BESS to enable these features. The Gantt chart and BESS usage are shown in Figures 5 and 6. As one can observe, the difference is minimal compared with the case in A2. Also, the BESS use follows the energy price, and the battery is fully charged in the beginning to be discharged during the most expensive times and toward the end. Note that here we focus on the pure energy amount in BESS and do not consider healthy BESS operating ranges (typically 10−90%), which would roughly require a 20% higher BESS capacity.

A BESS price higher than 300 kEUR/MWh would not have resulted in any investment decision, which shows that the added flexibility does, in this pricing scenario, not provide significant economic benefits. However, in operations, it could have importance during larger price spikes. It should be noticed that here we know the electricity prices shown in Table 1; thus, the decisions are based on ground truth or "perfect knowledge," which is not realistic. Also, such an investment decision cannot be made based on 2 days of electricity price information but requires a much more thorough analysis of the expected average values.

**3.3. Optimization of C-Cases by Enabling Bidirectional Trading.** The focus of this work is mainly to scratch the surface and compare different operational environments to get an indication of the potential of various options from which a large energy consumer can select. The simplest approach for bidirectional training is to allow the sales from BESS back to the grid, thus potentially in addition to balancing price spikes, benefiting from arbitrage trading,[50] i.e., buying when electricity is cheap and selling it back when the price is high. In this case, as seen in Table 4, the added benefit is small (around 680 EUR).

*3.3.1. More Complex Trading Options.* A more interesting option is to enable BESS for various trading options. This is a common approach for hybrid renewable power plants, e.g., wind parks, where a BESS unit is used to balance smaller fluctuations and ramp up/down the nondispatchable generation in a controlled way.[51] Processes are more and more

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Table 4. Enabling Bidirectional Energy Trading for the BESS

| BESS size (MWh) | profit | electricity cost | profit improvement |
|---|---|---|---|
| 44.5 (only sell) | 202,458.33 | 182,715.19 | 0.2% |
| 44.5 (buy/sell) | 203,138.86 | 182,034.66 | 0.5% |

expected to participate in balancing the grid through various grid products, and as the certification of a production process being accepted for reserve markets can be cumbersome, an existing BESS may be used to provide ancillary services. In the following, we assume the double role of a BESS, as depicted in Figure 1, supporting the process and trading energy with the power grid. Participating in electricity markets is complex, as there are various products (energy and ancillary services) as well as multiple markets (day-ahead, intraday, and imbalance), which can be bid and cleared at different times.[52] This means that auction-based markets collect the bids at a certain time of day (price and volume), and after the clearance, the bids are either accepted or rejected. Accepted bids must be considered when placing bids for the consecutive markets since the energy must be available to meet all committed bids. In practice, bidding for the first market of, e.g., tomorrow, must consider the forecasts for all other markets to be able to decide how to optimally distribute the capacity among the markets. In this paper, we assume having perfect forecasts and focus on the interaction between the different markets and estimate the maximum potential of the power grid interactions. There are, nonetheless, many other rules for most markets that complicate the bidding optimization. In some markets, you can only offer a given number of price levels (price band) per day and often the bid volume must be a discrete number, i.e., 0.1 or 1 MW, which requires additional modeling. Here, we do the optimization iteratively in the order of bidding timelines as shown in Figure 7. Furthermore, as ancillary events are very random and typically occur at 1−2% of the time points, the possible BESS shortage as a result of successive events is not considered in this context of the bidding phase. Often, grid operators, knowing the BESS limitations, ensure a good mix of capacities for each hour. The market model can also build a certain tolerance, e.g., by limiting bid volumes to tolerate two or more successive events. Methods to "recover" from an unplanned state can be done through balancing markets or locally by potentially adapting the production schedule.

The loop is similar to a simulation run. We start by simultaneously focusing on all products and markets, create the price bands (if needed) based on the available forecasts, optimize the entire bids maximizing a profit function, and then place the bids for the next market to be cleared. As soon as the market has been cleared ("real-life interaction" in Figure 7), the accepted bids are committed and fixed, and the same procedure is repeated for the other products and markets. Often, the forecasts may change between the runs, so it is important always to use the most recent and accurate forecast available. In energy markets, energy transactions are realized according to the market clearance. For reserve markets, e.g., ancillary services, the energy must be kept in reserve, and actual transactions only take place based on the need, i.e., participants are remunerated based on their readiness to act. This can be attractive, especially in environments where ancillary events are rare, which basically generate revenues without any action. Depending on the price structures, the optimization can then decide between trading energy or reserving the BESS capacity for ancillary services.

The most important issue is to ensure that the energy balance is respected, i.e., that we do not buy more than we consume and can store in the battery or sell more than is available at the battery storage. The corresponding equation is shown in eq 9:

$$x_t^{\text{buy}} + \sum_{B\in\text{BESS}} x_{B,t}^{\text{dc}}\cdot\Delta_t = X_t^{\text{load}}\cdot\Delta_t + x_t^{\text{sell}} + \sum_{B\in\text{BESS}} x_{B,t}^{\text{ch}}\cdot\Delta_t \quad \forall\, t \tag{9}$$

The amount of energy bought, $x_t^{\text{buy}}$, and discharged from the batteries, $x_{B,t}^{\text{dc}}\cdot\Delta_t$, must at all times equal the load, $X_t^{\text{load}}\cdot\Delta_t$, plus energy sold, $x_t^{\text{sell}}$, and energy charged, $x_{B,t}^{\text{ch}}\cdot\Delta_t$. We multiply the load and dis/charging variables by the time grid $\Delta_t$ because they are measured by power (MW), whereas other terms refer to energy (MWh). The earlier defined RTN-based process model is connected to the energy balance via eq 10.

$$\pi_{\text{elec},t} = X_t^{\text{load}} \quad \forall\, t \tag{10}$$

With these equations, we can elegantly link the energy market transactions with the studied RTN-based process model. This allows us to directly link very complex bidding models to the variable load of the generated RTN schedule. It also allows for a natural expansion of the trading model to consider electricity price uncertainty without compromising the process model. We assume a BESS configuration with the capacity of 44.5 MWh, and an initial state of charge (SoC) of 0% (0 MWh), which can be fully (dis)charged in 2 h. In this study, we do not take the typical 50% initial capacity assumption, as it could provide too high a benefit in this short time frame. In addition, we assume a discharging efficiency of 95% and also penalize every discharging operation to account for battery degradation.

Bidding into different electricity markets can be with or without a BESS.[50,51] The focus of most industrial DSM studies has been on the energy market. Typically, large consumers have multiple electricity contracts, e.g., time-of-use or fixed-base tariffs, until a certain threshold. To avoid further complexities, we assume here that our example process will

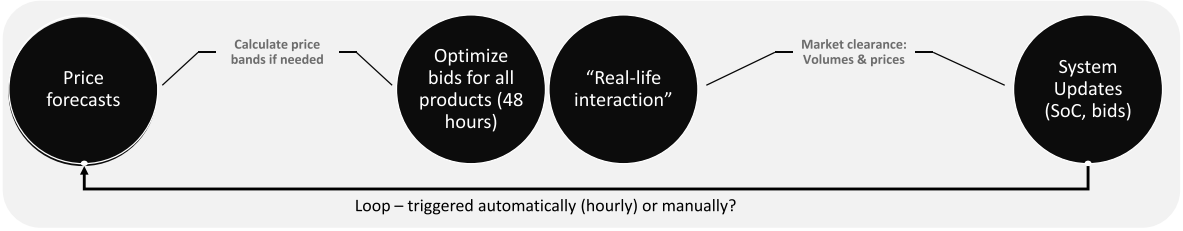



Figure 7. Optimization done in a loop following the bidding sequence.

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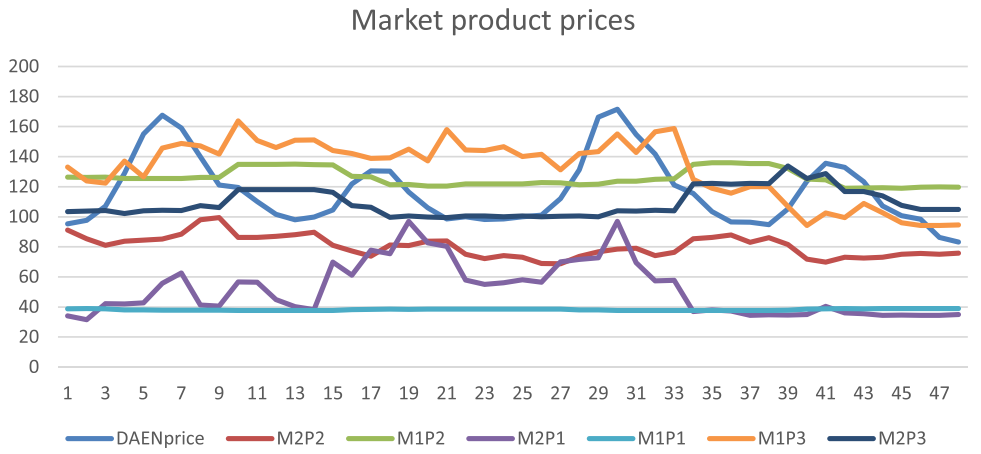



**Figure 8.** Prices of considered products and markets (EUR/MWh for energy, EUR/MW for ancillary services).

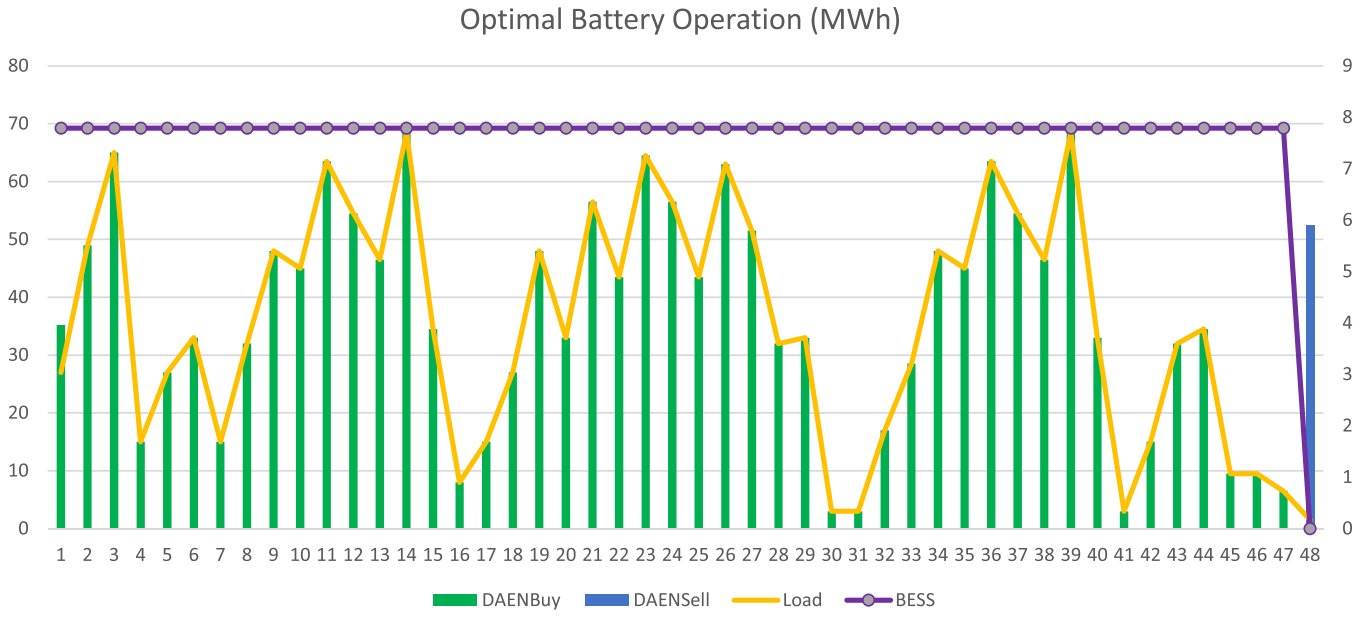



**Figure 9.** Energy purchase, load (left axis), energy sales, and BESS state of charge (right axis).

not impact the market-clearing price and acts as a "price taker." However, for very large consumers, this may not be true. From the power grid perspective, it is becoming increasingly important that large consumers also bid into ancillary service markets and help stabilize the power grid during network disturbances, i.e., frequency drops below a given threshold. As committed ancillary services are always required to be available, even if mostly not used, the most secure and common approach is to store the energy within a BESS. The bidding part of this paper is based on Hafiz et al.,[51] which comprises a solar power plant combined with a BESS. Here, we focus on the process system, ignoring the renewable production.

*3.3.2. Markets and Products.* Apart from the energy market discussed earlier, we consider two different ancillary markets, where market M1 requires bidding 24 h before the start of the trading day and market M2, respectively, 12 h before. The ancillary products comprise both up- and down-regulation of the frequency through either injecting (discharging BESS) or extracting (charging BESS) power to/from the grid. We name the products as follows: P1 (up), P2 (down), and P3 (up/down). Product P3 is symmetric and must be able to provide exactly the same potential capacity (MW) in both directions. The ancillary service capacity cannot be shared with multiple products. For energy trading, we use only the day-ahead energy market (DAEN). The prices used in this example are shown in Figure 8 and, as before, they are extracted from the DK-2 spot market (June 28−29, 2023).

As can be seen from the price curves, the various products do not follow the same trends, which provides an opportunity to use bidding as a means of optimizing the portfolio among the various products. An interesting fact is that energy is bid in MWh, whereas ancillary service bids are placed in MW. As some ancillary services (here, P1 and P2) are required only to provide services for 20 min, an offer of 18 MW for a given trading hour maximally requires only 6 MWh of energy if called upon. In the modeling, one must keep track of this power-to-energy (P2E) factor in order to ensure that the BESS SoC is sufficient for the offered services. The factor P2E has the unit hours multiplied by the time the power must be delivered and divided by the length of an hour. The ancillary product P3 must be able to deliver for the full hour (P2E = 1 h). This also means that in case there is a limitation of the battery dis/charging rate (MW) or bus transaction rate, the

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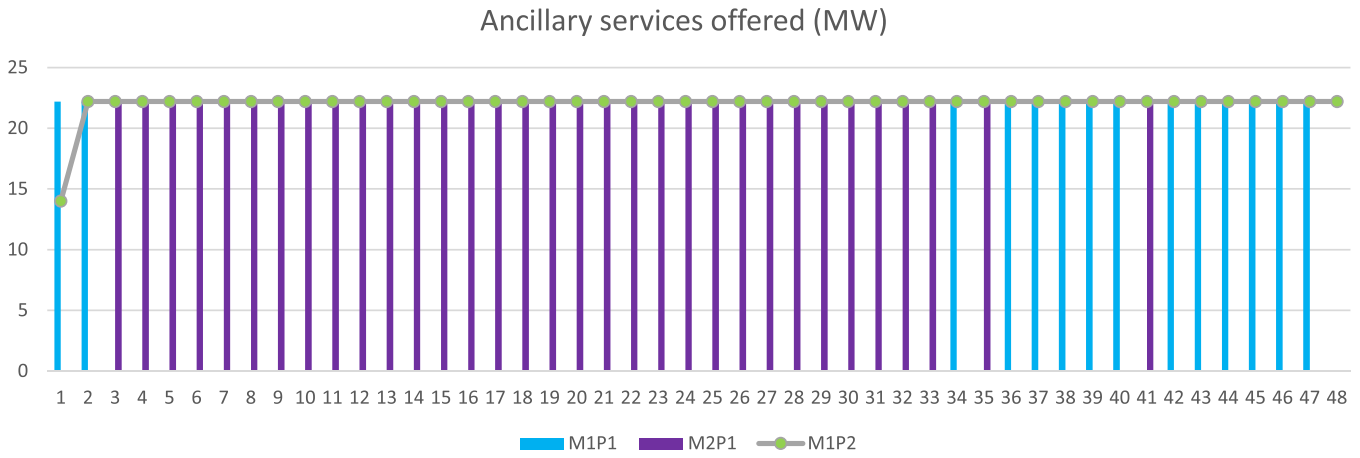



**Figure 10.** Ancillary operations showing that positive reserves (P1) alternate, whereas negative reserves (P2) are stable.

momentary power must not exceed this limit at any time. As a concrete example, the BESS of 44.5 MWh can, at maximum, inject or receive 22.25 MW, and ancillary services may only be provided by the energy present in the BESS. Thus, while discharging the battery by 10 MW for either providing energy to the process or selling electricity back to the grid, we can offer a maximum of 12.25 MW of ancillary services as a positive reserve.

The results of the optimization run using these energy market options are shown in Figure 9, where it can be seen that in the beginning slightly more energy is purchased as the load to raise the BESS SoC to 7.79 MWh (right axis), keeping it constant until the end, where the energy is sold back to the grid (5.9 MWh) to deplete the BESS. In all other time points, the energy purchase equals the load. Thus, here, we assume an empty BESS in the beginning to avoid unfair benefits compared to having no BESS available. For the same reason, the BESS is emptied at the end of the horizon. In normal continuous operation, this would not be the case, as typically the SoC at the end is nonzero and equals the initial one.

This means that the BESS is fully used for ancillary services, which, in this ideal setup, without ancillary events, is optimal. In practice, with the existing market price information, the BESS is treated as a decoupled entity and does not contribute to or impact the steel production schedule. In Figure 10, we can see the optimal ancillary services. The product P1 (positive reserve) is alternated between the two markets, always selecting the most expensive one, and the same capacity is offered for P2 (negative reserve). Both products are bid by 22.2 MW (maximum dis/charging capacity) and coincide with the needed energy amount of 7.79 MWh available at BESS, multiplied by the efficiency factor 95% and divided by PoE = 1/3 (20 min provision per hourly bid).

Thus, all available energy is fully used to maximize the profit, which is now increased to 384 kEUR, showing an increase of 90%. This comes from the ancillary sales income, which is 188,853 EUR vs 192,025 EUR for electricity purchases, pushing down the total electricity cost to 3.2 kEUR, as shown in Table 5.

The actual production schedule is shown in Figure 11. The example results show that, in an ideal situation, using an available BESS can help to level the energy costs but is not necessarily used for smoothing the production schedule. In all fairness, here, we had relaxed conditions and, e.g., did not penalize for longer intermediate residence times but only restricted the amount of each intermediate state to 25 tons (one batch). The current model shows the capabilities for supporting short-term operations but cannot be used for CAPEX decisions. However, it very well illustrates the elasticity of adapting the decisions based on changing data, something where mathematical optimization approaches are superior if the models are linear and the combinatorial space is manageable.

**Table 5. Results from Including the Ancillary Services Enabled by the BESS**

| BESS size (MWh) | profit | electricity cost | profit improvement |
|---|---|---|---|
| 44.5 (w/ancillary) | 384,507.46 | 3,171.55 | 90.3% |

## 4. VARYING FLEXIBILITY AND PRICE SCENARIOS

This paper has used an illustrative example along with the model development to discuss the different levels of integrating a production process into the energy markets (options A−C). It has been shown that we can together model the process and related energy storage and trading. The selected RTN-based approach can very well be used for short-term planning (a few days) of an energy-intensive process. However, the examples are very specific and unrealistic in a few ways:

- We have only used energy prices of two selected days.
- We assume that the energy price forecasts are 100% accurate.
- We do not experience any ancillary events—even if reserving capacity for those.
- The process we use as an example has minimum flexibility (MS 47 in a planning horizon of 48 h).
- Process simplifications make it possible to do some shifting, which enables some potential to adapt to the electricity prices.

In this paper, we will not consider all those aspects but just illustrate some of the effects by using different data and compare the modified results to the above ones. In short, we will first shift the energy price days and compare the outcome. Then, we also reduce the production target, and instead of 21 batches, we only need to produce 18 batches and look how much of the added flexibility is utilized when co-optimizing the energy costs. The main idea here is just to show the need for solving larger and well-selected instances on the basis of longer-term decision-making. As mentioned above, here, we

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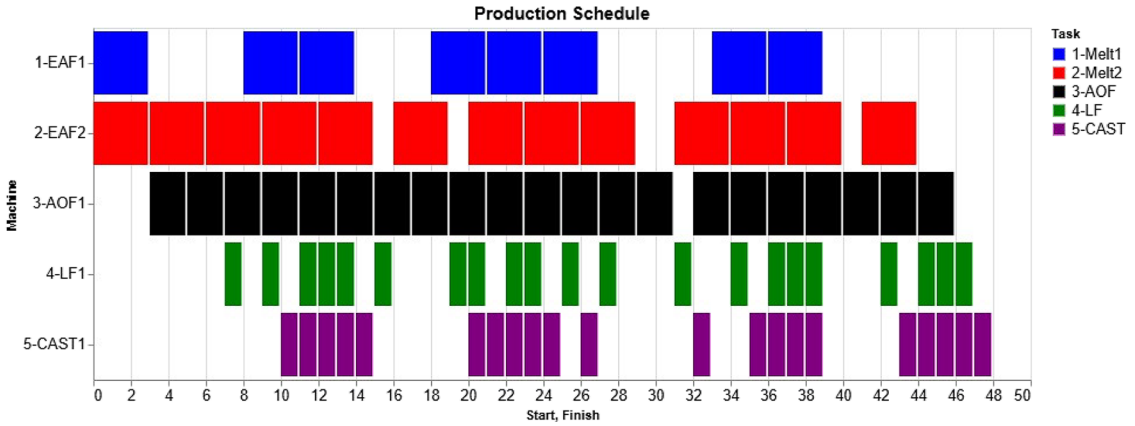



**Figure 11.** Gantt chart using a BESS of 44.5 MWh when bidding for ancillary services (time unit: hours).

**Table 6. Electricity Price (EUR/MWh) for 30.6.−1.7.2023**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 76.94 | 77.36 | 84.31 | 107.09 | 115.13 | 131.52 | 130.74 | 122.39 | 109.47 | 109.74 | 99.50 | 85.48 | 79.96 | 67.30 | 53.64 | 54.27 | 55.12 | 76.09 | 64.90 | 43.79 | 24.61 | 16.83 | 4.43 | 0.07 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 0.99 | 12.67 | 52.40 | 67.88 | 67.27 | 55.87 | 55.57 | 55.12 | 45.34 | 16.45 | 3.17 | 0.01 | 0.00 | -0.03 | -0.05 | -0.05 | -0.50 | -2.95 | -5.03 | -4.24 | -3.07 | -2.50 | -4.21 | -6.62 |

assume perfect forecasts. These are linked to the problem through the market model, which takes as inputs a set of selected scenarios. In these cases, we assume a single scenario with 100% probability (perfect forecast), as shown in Table 1, and based on true historical market-clearing prices. The same market model can equally well use ML-driven forecasts and generate scenarios across selected quantiles.

**4.1. Electricity Pricing.** In the following, we shift the energy prices forward by 2 days. The energy prices in Table 6 are again taken from the DK-2 spot market (June 30−July 1, 2023), which shows much more volatility and even negative prices toward the end, with an average price going down from 116 EUR/MWh to 45 EUR/MWh. This will obviously have a positive effect on the profit owing to the lower electricity costs.

## 5. RESULTS AND DISCUSSION

The earlier cases have been solved and shown along with the method description, and the corresponding results are presented in Tables 3−5. To create room for discussion, first, we solve all the cases from before with only the changed electricity prices and shortly discuss the results. All of the runs are shown in Table 7. In the first cases, the lowered electricity costs contribute to higher profits, but at the same time, the effect of a BESS is lower than above due to the same reason. Having a 50 MWh BESS would only increase the profit by slightly more than 2%. Assuming free and unbounded battery storage capacity, the optimization would here settle for 438.5 MWh, which is mainly used to purchase electricity during the lowest (even negative) electricity prices.

**Table 7. Cost Re-Optimization with Different BESS Capacities and Electricity Prices**

| BESS size (MWh) | profit | electricity cost | profit improvement |
|---|---|---|---|
| 0 (case A1) | 306,364.60 | 87,385.41 | NA |
| 0 (case A2) | 321,052.87 | 72,697.13 | 4.8% |
| 10 | 322,525.28 | 71,224.72 | 0.46% |
| 20 | 323,963.79 | 69,786.21 | 0.91% |
| 30 | 325,310.89 | 68,439.11 | 1.33% |
| 40 | 326,587.47 | 67,162.53 | 1.72% |
| 50 | 327,706.48 | 66,043.52 | 2.07% |
| 438.5 (free) | 343,803.48 | 49,946.52 | 7.09% |
| 0 (incl. CAPEX) | 321,052.87 | 72,697.13 | 0% |
| 0 enable selling | 321,052.87 | 72,697.13 | 0% |
| 44.5 (w/ancillary) | 639,277.38 | −254,103.86 | 99.1% |

Even if just a BESS for increasing the flexibility in energy purchasing does not provide a high benefit, in this case, the bidding for ancillary services is much more lucrative. This is due to the fact that during low electricity pricing, which typically means a very high renewable energy share, the prices for ancillary services lowering the frequency (P2) go up, and the symmetric ancillary product P3 even reaches prices around 850 EUR/MW, as seen in Figure 12. This also has an impact on the SoC levels, enabling the participation in these services, which can also be reflected in the production schedule shown in Figure 13.

The response to the price changes is best shown in the resulting energy strategy. In Figure 14, it can be seen how the BESS SoC is raised to the 7.79 MWh level immediately in the beginning to be able to offer the most valuable services, and later when the P3 price increases, the BESS is adapted to provide the maximum service (P3 is symmetric and requires full-hour delivery) exactly at the middle point of its capacity. The services P2 and P3 alternate, as seen in Figure 15, based on their price levels.

**5.1. Capacity Utilization vs Flexibility.** Another important factor that must be considered in the context of DSM is the flexibility that a process can offer, which is key to enabling variations in energy purchases. In the previous case, we have a very congested process where the minimum makespan is 47 h within a 48 h planning horizon. This alone offers very little flexibility, as was seen in testing the various BESS sizes, and therefore, the ancillary service opportunity was used. However, the model could also switch to alter the energy trading if the price structures favor this. As a last sensitivity analysis topic, we will reduce the production target to 450 t (18 batches) and redo the first cases, the results of which are shown in Table 8.

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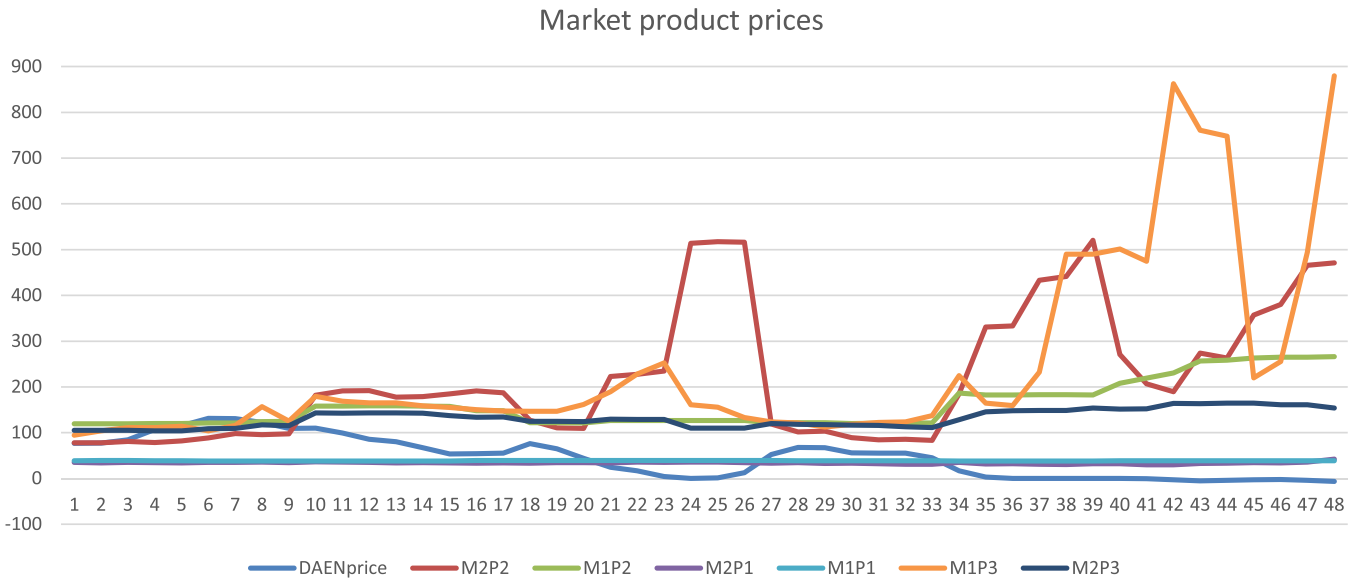



**Figure 12.** Prices of considered products and markets for the second studied time period (EUR/MWh for energy and EUR/MW for ancillary services).

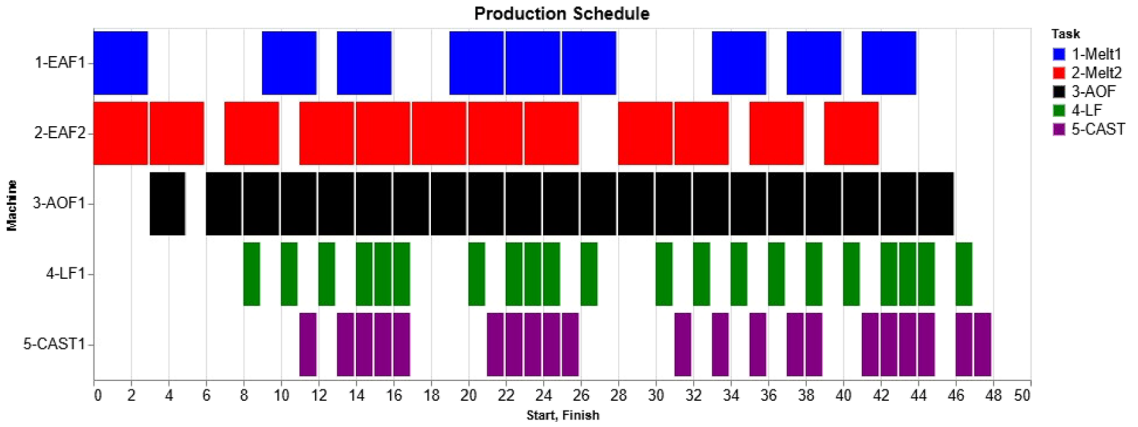



**Figure 13.** Production schedule using 44.5 MWh BESS and including ancillary services (time unit: hours).

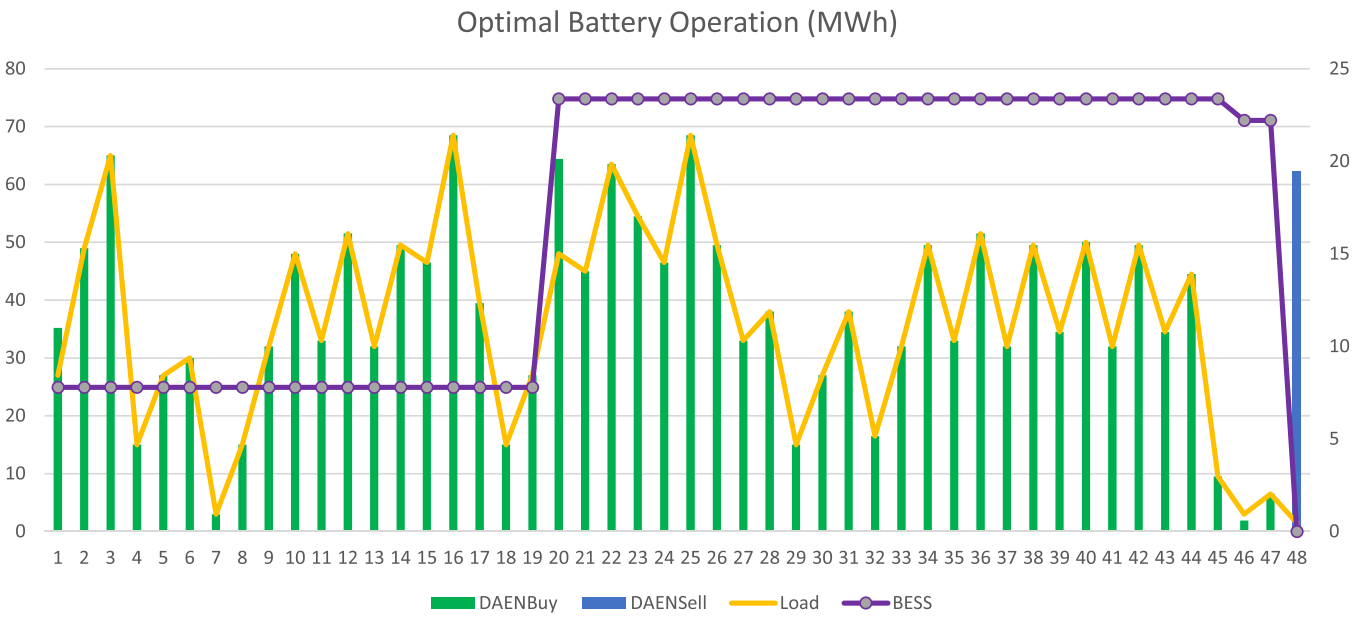



**Figure 14.** Energy trading including the process load and the 44.5 MWh BESS stage of charge (right axis).



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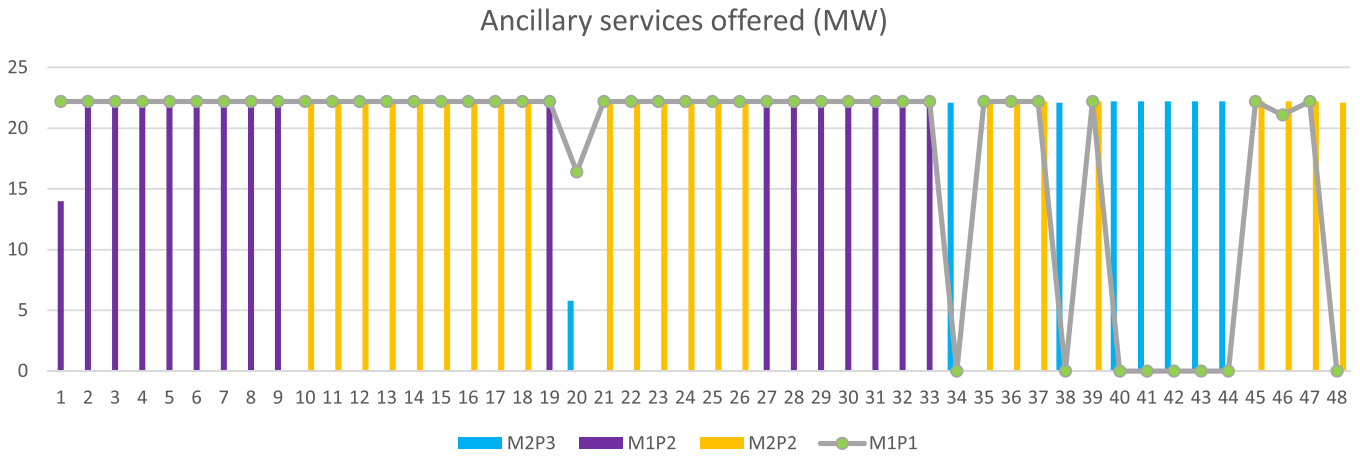



**Figure 15.** Ancillary service strategy with 44.5 MWh BESS.

**Table 8. Cost Optimization with 450 Ton Production Target**

| BESS size (MWh) | profit | electricity cost | profit improvement |
|---|---|---|---|
| 0 (case A1) | 156,591.04 | 180,908.96 | NA |
| 0 (case A2) | 180,741.00 | 156,759.00 | 15.4% |
| 10 | 182,329.22 | 155,170.78 | 0.88% |
| 20 | 183,558.68 | 153,941.33 | 1.56% |
| 30 | 184,585.37 | 152,914.63 | 2.13% |
| 40 | 185,618.67 | 151,881.33 | 2.7% |
| 50 | 186,163.13 | 151,336.87 | 3.0% |
| 1111.0 (free) | 199,551.45 | 137,948.55 | 10.4% |
| 3 (incl. CAPEX) | 180,770.46 | 156,151.35 | 0.02% |
| 3 enable selling | 180,823.77 | 156,098.04 | 0.37% |
| 3 (w/ancillary) | 184,711.65 | 144,211.87 | 2.2% |

Here, the most significant change can be seen between cases A1 and A2, i.e. if there is a flexibility opportunity (unused capacity which can be shifted), it is essential to add the electricity costs into the objective. Due to the already existing flexibility, the additional BESS capacity does not significantly lower the electricity cost. Therefore, the optimization including CAPEX only suggests a 3 MWh battery, and using it for ancillary services helps to lower the electricity cost by about 12 kEUR.

## 6. CONCLUSIONS

In this paper, we have discussed enhancing the traditional DSM, where energy-intensive operations are adapted based on the forecasted or known electricity prices. The integration of the BESS capacity with the production process is a central component of the proposed approach. This could help decouple the energy purchases from the actual momentary demand.

In the first part, we investigated the effect of selected fixed battery sizes, and the benefits were smaller than expected but still within a few percent. Including the investment cost made it barely profitable, especially if the BESS price is above 250 kEUR/MWh. However, it must again be highlighted that such an investment decision should not be made based on a few days' price forecasts.

The main potential was, nonetheless, discovered when considering the electricity price in the production scheduling optimization, which offered 5−15% savings depending on the process flexibility to adapt its production accordingly. Even if the investment was not sufficiently motivated by the traditional DSM perspective, a BESS can also be used for ancillary trading—either partially in parallel or entirely to support the grid stability. It can be expected that in the future, an increased contribution to the reserve markets will be needed from the industry. Here, potential benefits turned out to be significant when operating in an ideal environment, assuming perfect knowledge and that all bids are accepted and no ancillary events occur. The income from the ancillary services could even fully cover the energy costs.

Thus, we have seen that theoretically there is a potential, and there are still many avenues that require further research before the positive effect can be proven. Some of them are listed below.

- Process flexibility: The production targets and product quality must not be endangered due to electricity cost minimization—here, a stronger BESS-process integration can be essential.
- Price uncertainty: As the energy prices are expected to fluctuate even more in the future, accurate forecasting methods are needed[53] to enable better and more efficient actions to minimize the energy cost.
- If the forecasting quality is not sufficiently high, methods dealing with the price uncertainty and stochasticity must be brought to a level that they can be used also for short-term operational planning.
- Bidding strategies could be developed to alter between the "traditional DSM," i.e., energy trading and potentially more profitable ancillary markets that were discussed in this paper, including the consideration of realistic ancillary events.
- As current electricity markets are highly structured for generation-type participants, new policies and rules must be included to also ensure the operational feasibility for industrial participants, in order to enable the participation of more complex industrial loads.
- Control: Should the grid operator control all or part of the load? How can an industrial user offer different levels of flexibility, with respect to response time and duration of its load change?

The main challenge remains how to practically and economically realize the true integration of large industrial loads with the energy markets. It is in everyone's common interest to maintain a stable and healthy energy infrastructure.

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DSM opportunities should be financially attractive to motivate investments in more modern and flexible production systems. Here, potential cross-site or even cross-company collaboration may be deployed to lower individual risks.

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### Notes

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